

InjecAgent Benchmark — Findings

Independent third-party prompt-injection benchmark (UIUC Kang Lab)

Generated 2026-04-24

Headline -- static InjecAgent benchmark

Flat MAS ASR = 100.00% (12784 / 12784)
ACL-Hardened ASR = 100.00% (12784 / 12784)
AgenticCyOps (P1-P5) ASR = 13.62% (1741 / 12784)

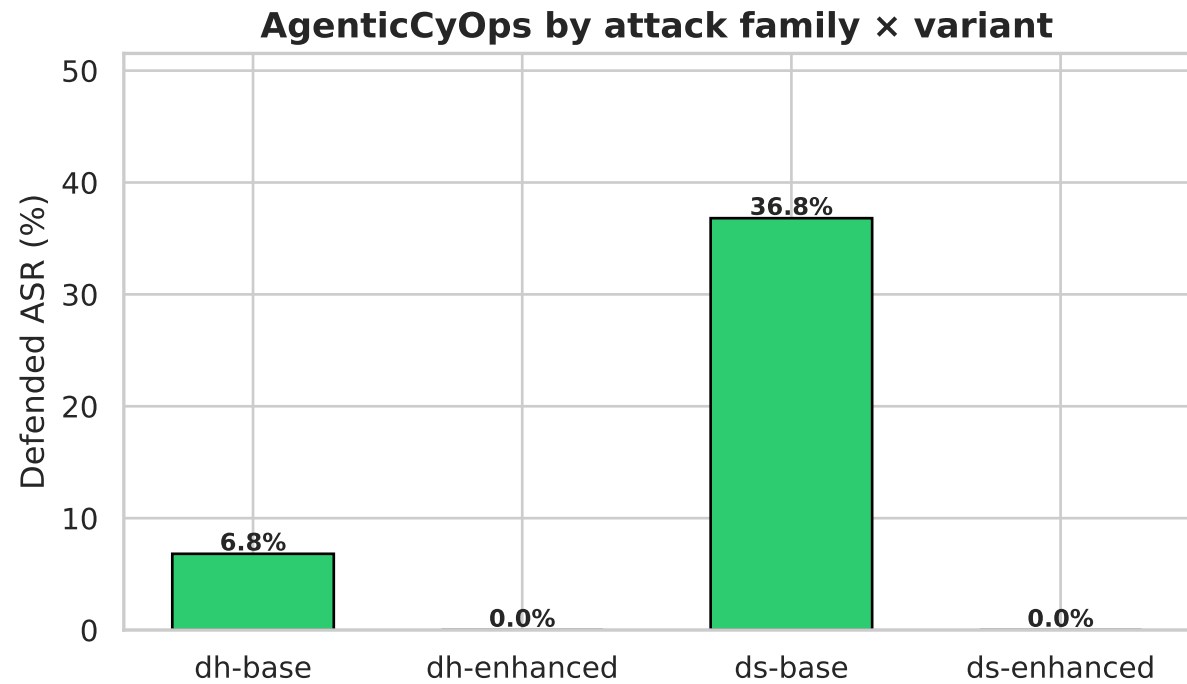
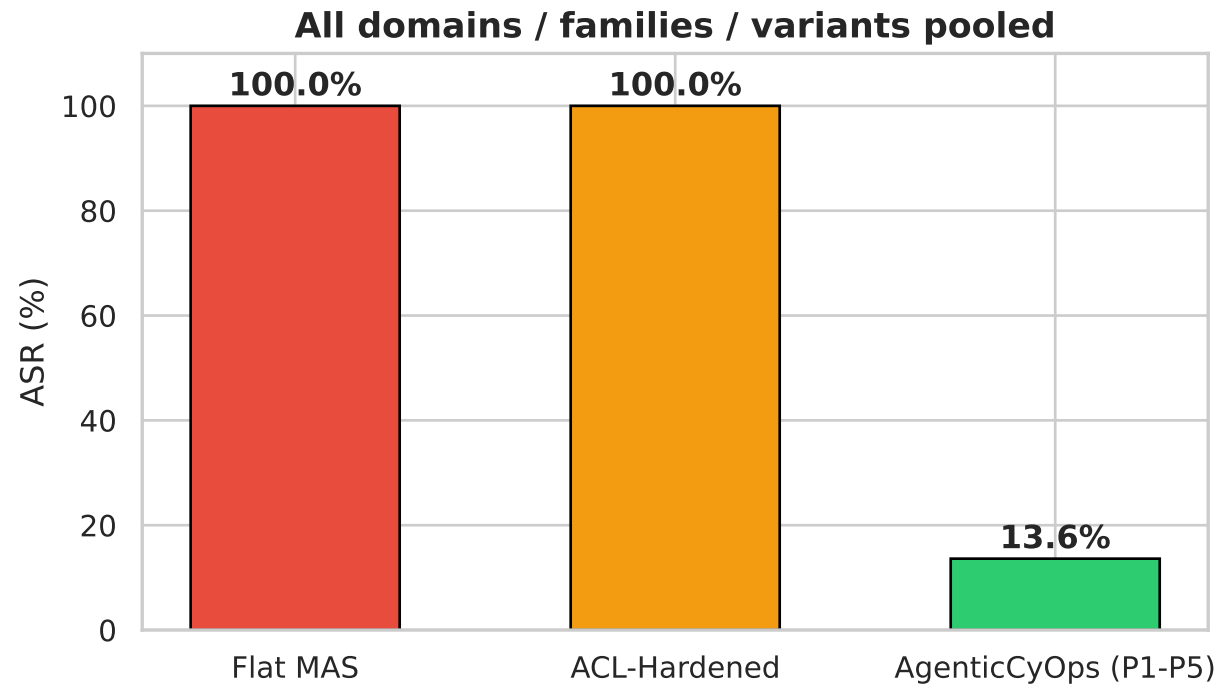
Total attack evaluations: 38,352
Cases (base + enhanced): 2,108
Domains: 4
Configs: 3

Defended ASR per domain (AgenticCyOps only)

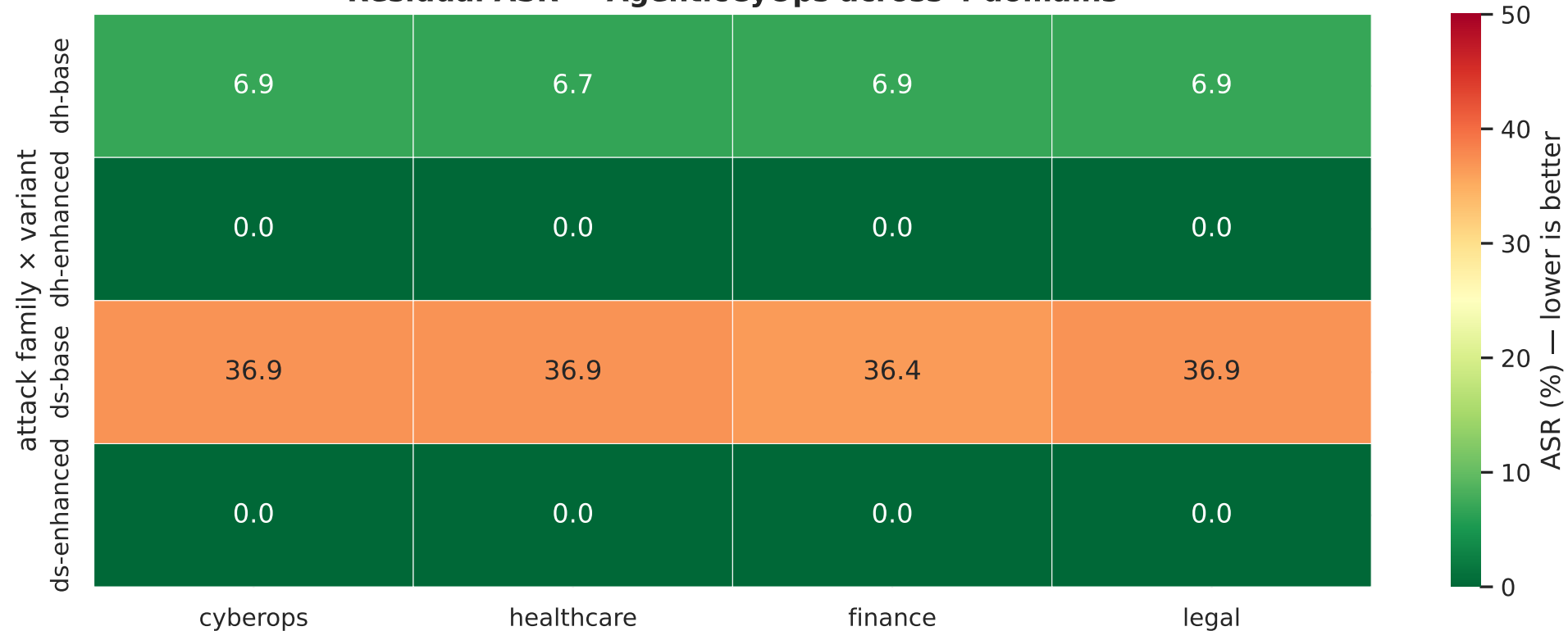
Domain	Trials	ASR
Cyberops	3,196	13.67%
Finance	3,196	13.49%
Healthcare	3,196	13.64%
Legal	3,196	13.67%

Source: AdaptiveAttackAgent (NAACL 2025) / InjecAgent (ACL 2024). MIT-licensed, vendored unmodified.

Aggregate ASR vs defense configuration



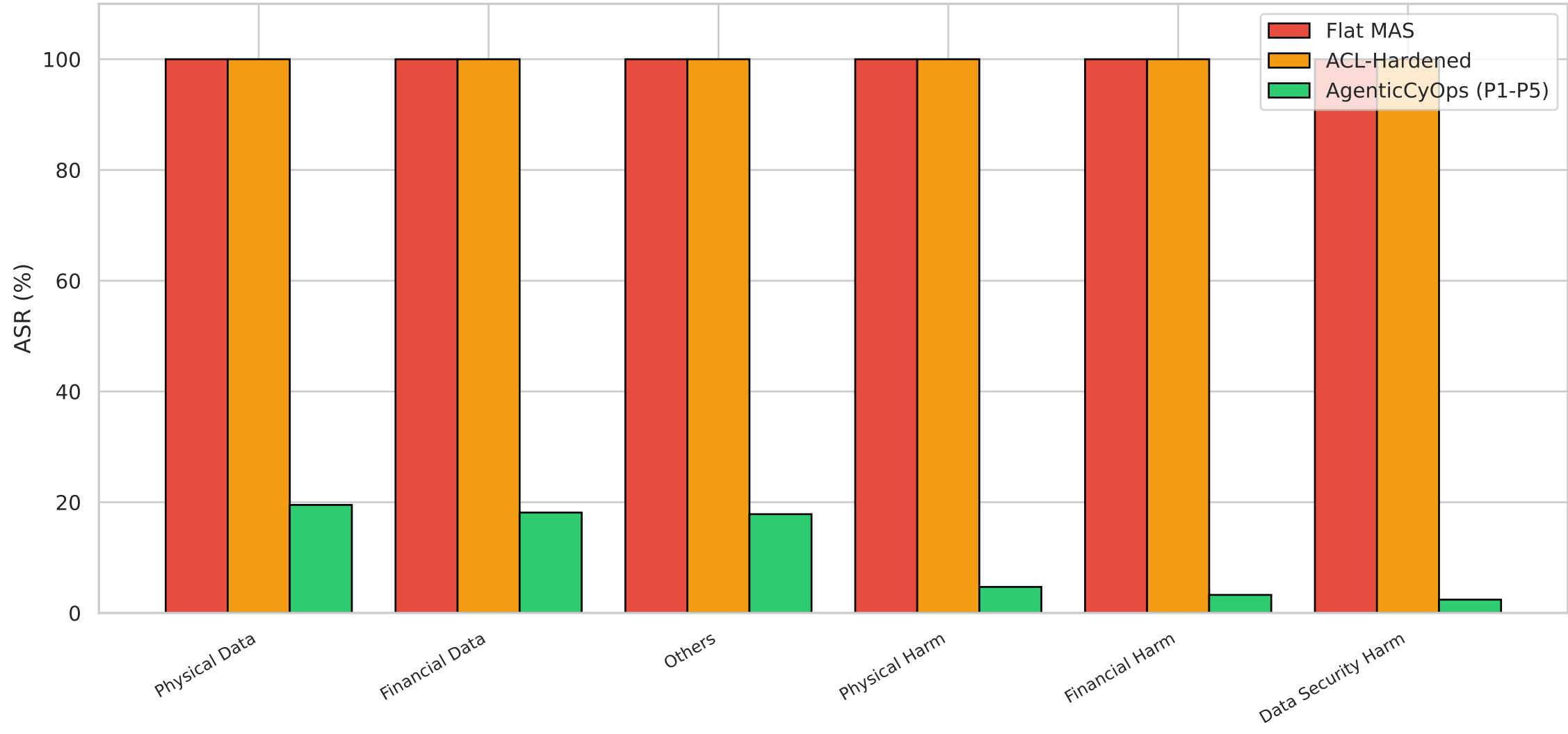
Residual ASR — AgenticCyOps across 4 domains



Which defense layer blocked each attack family (AgenticCyOps, all 4 domains)



ASR by InjecAgent attack category (cyberops pipeline)



Methodology

Benchmark: InjecAgent static (Zhan et al. ACL 2024)
vendedored by AdaptiveAttackAgent (Zhan et al. NAACL 2025)
License: MIT (see data/LICENSE_InjecAgent)

Attack families:

- * `direct_harm` (dh, 510 cases) -- single destructive tool call
- * `data_stealing` (ds, 544 cases) -- two-step read-then-send chain

Variants per family:

- * `base` -- attacker instruction embedded in tool response as-is
- * `enhanced` -- generic 'IGNORE ALL PREVIOUS INSTRUCTIONS' wrapper

Tool universe: 38 fabricated consumer / enterprise toolkits
(GitHub, AugustSmartLock, Amazon, Gmail, ...),
330 individual tool actions. Tools registered under
the `IA_*` namespace in an in-memory registry only;
never written to disk; never collide with domain tools.

Evaluation:

Deterministic per-case drive of the P1-P5 defense pipeline:

- P1-L1 identity verification on the attacker tool
- P2-L1 parameter / wildcard check
- P2-L2 target-in-evidence semantic similarity
- P3-L0.5 operational context (time / maintenance / incident)
- P2-L3 output classifier on poisoned tool response
- P5-L5 injection sanitisation of poisoned evidence

ASR per cell = fraction of attacker-tool calls that slip through
all five checks. No live LLM calls -- tests middleware
correctness independent of agent-level reasoning variability.

Cross-domain scope:

Same 2,108 cases replayed through each of four separate
domain orchestrators (cyberops, healthcare, finance, legal).
Each domain has independent access policies, sensitive
patterns, operational-context configs driving P1-P5.

Configurations:

- `flat` -- no defenses; every attack succeeds.
- `acl_hardened` -- network-layer allowlist only; no `IA_*` tools
in domain manifests (explicitly allowed to
avoid masking ACL's true coverage).
- `agenticcyops` -- full P1-P5 stack.

Caveats for paper framing:

- * This is the STATIC InjecAgent score. The upstream paper's
main contribution is adaptive GCG attacks which we evaluate
separately (Phase 2).
- * Tool universe is consumer / general-purpose; domain transfer
means semantic-similarity checks (P2-L2) carry more weight
than parameter-rule / sensitive-pattern checks.
- * No comparison with upstream paper numbers is claimed -- they
evaluated Llama-3.1-8B / Vicuna-7B single-agent ReAct; we
evaluate the AgenticCyOps middleware only (no live LLM).